

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Date:

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

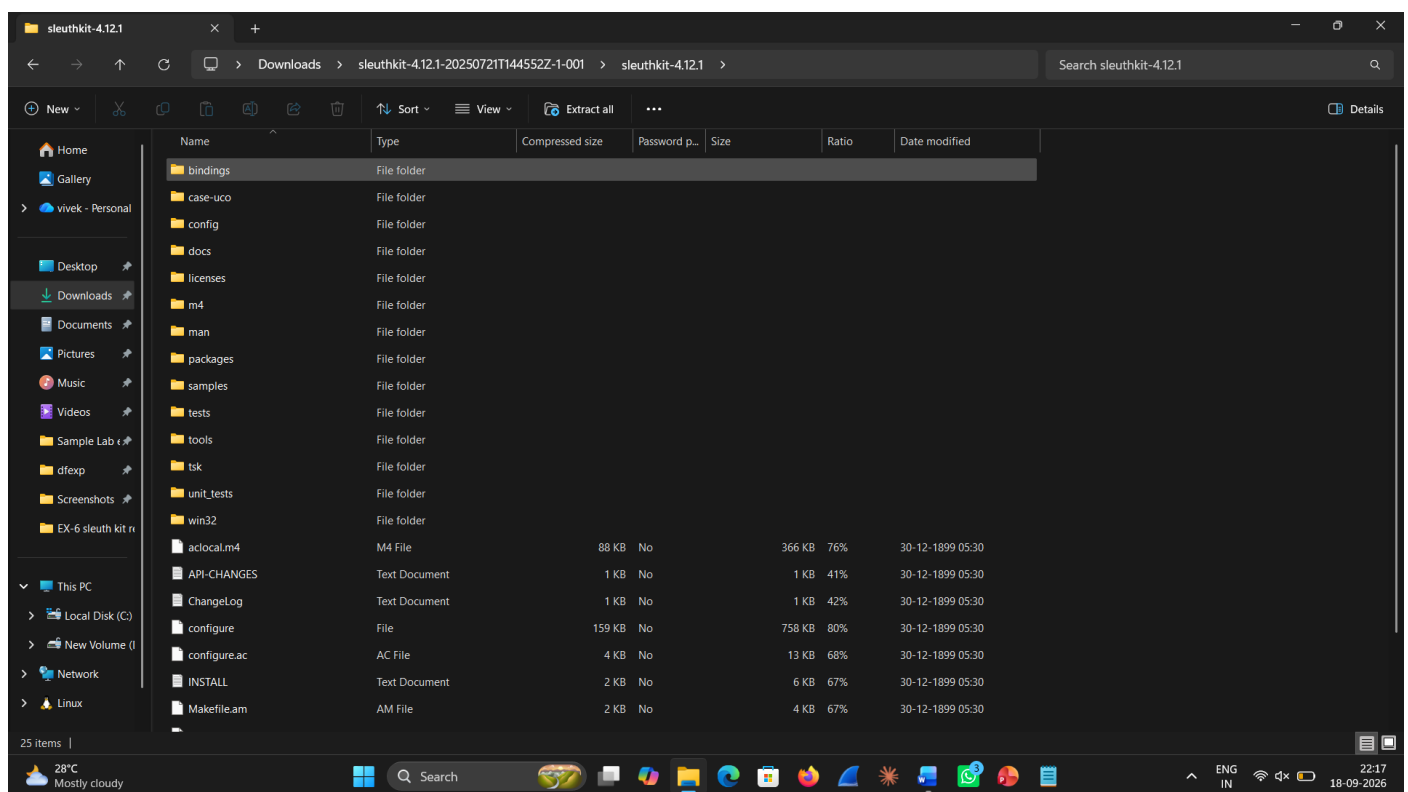
Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.



Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

 4Dell Latitude CPi (1).E01	17-09-2026 16:09	E01 File	6,55,366 KB
 4Dell Latitude CPi (1).E02	17-09-2026 16:09	E02 File	4,09,556 KB

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

```
arduino
```

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command:

```
arduino
```

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

```
css
```

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

```
css
```

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

```
Windows PowerShell
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> .\fsstat.exe --help
Invalid argument: (null)
usage: C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin\fsstat.exe [-tvV] [-f fstype] [-i imdtype] [-b dev_sector_size] [-o imgoff
set] image
    -t, display type only
    -i imdtype: The format of the image file (use '-i list' for supported types)
    -b dev_sector_size: The size (in bytes) of the device sectors
    -f fstype: File system type (use '-f list' for supported types)
    -o imgoffset: The offset of the file system in the image (in sectors)
    -P pooltype: Pool container type (use '-P list' for supported types)
    -B pool_volume_block: Starting block (for pool volumes only)
    -v: verbose output to stderr
    -V: Print version
    -k password: Encryption password for encrypted volumes
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> .\fsstat.exe "C:\Users\Anand\Downloads\Dell Latitude CPi (1).E01"
Cannot determine file system type
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> .\img_stat.exe "C:\Users\Anand\Downloads\Dell Latitude CPi (1).E01"
IMAGE FILE INFORMATION
-----
Image Type:                ewf
Size of data in bytes:    4871301120
Sector size:              512
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> Copy-Item "C:\Users\Anand\Downloads\Dell Latitude CPi.E02" "C:\Users\Anand\Down
loads\Dell Latitude CPi (1).E02"
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> Get-ChildItem "C:\Users\Anand\Downloads\Dell Latitude CPi*.E0?"

Directory: C:\Users\Anand\Downloads

Mode                LastWriteTime         Length Name
----                -
-a----          17-09-2026   16:09         671094597 4Dell Latitude CPi (1).E01
-a----          17-09-2026   16:09         419384951 4Dell Latitude CPi (1).E02
-a----          30-08-2026   21:56         671094597 4Dell Latitude CPi.E01
-a----          17-09-2026   16:09         419384951 4Dell Latitude CPi.E02
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> .\img_stat.exe "C:\Users\Anand\Downloads\Dell Latitude CPi (1).E01"
```

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

CSS

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

CSS

```
mactime -b body.txt > timeline.txt
```

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

```
Windows PowerShell
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> .\fsstat.exe -o 63 "C:\Users\Anand\Downloads\Dell Latitude CPi (1).E01" > fi
lesystem_info.txt
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> Get-Item .\filesystem_info.txt

Directory: C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin

Mode                LastWriteTime         Length Name
----                -
-a----          17-09-2026   16:38          2912 filesystem_info.txt
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> .\fls.exe -o 63 -r "C:\Users\Anand\Downloads\Dell Latitude CPi (1).E01" > fi
le_list.txt
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> Get-Item .\file_list.txt

Directory: C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin

Mode                LastWriteTime         Length Name
----                -
-a----          17-09-2026   16:39         853158 file_list.txt
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> Get-Content .\file_list.txt -TotalCount 40
d/d 3671-144-7: Documents and Settings
+ d/d 3673-144-7: All Users
++ d/d 3707-144-7: Application Data
+++ d/d 3800-144-7: desktop.ini
+++ d/d 3708-144-7: Microsoft
++++ d/d 3709-144-1: Crypto
+++++ d/d 3712-144-1: Dss
++++++ d/d 3713-144-1: MachineKeys
+++++ d/d 3710-144-1: RSA
+++++ d/d 3711-144-1: MachineKeys
+++++ d/d 10977-144-1: Dr Watson
+++++ f/r 10979-128-3: drwtsn32.log
+++++ f/r 10980-128-3: user.dmp
+++++ d/d 7325-144-1: HTML Help
```

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

```
PS C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> Get-ChildItem "C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32" -Recurse -Filter "mactime*"

Directory: C:\Users\Anand\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin

Mode                LastWriteTime         Length Name
----                -
-a-----         17-09-2026   16:18           27249 mactime.pl
```

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

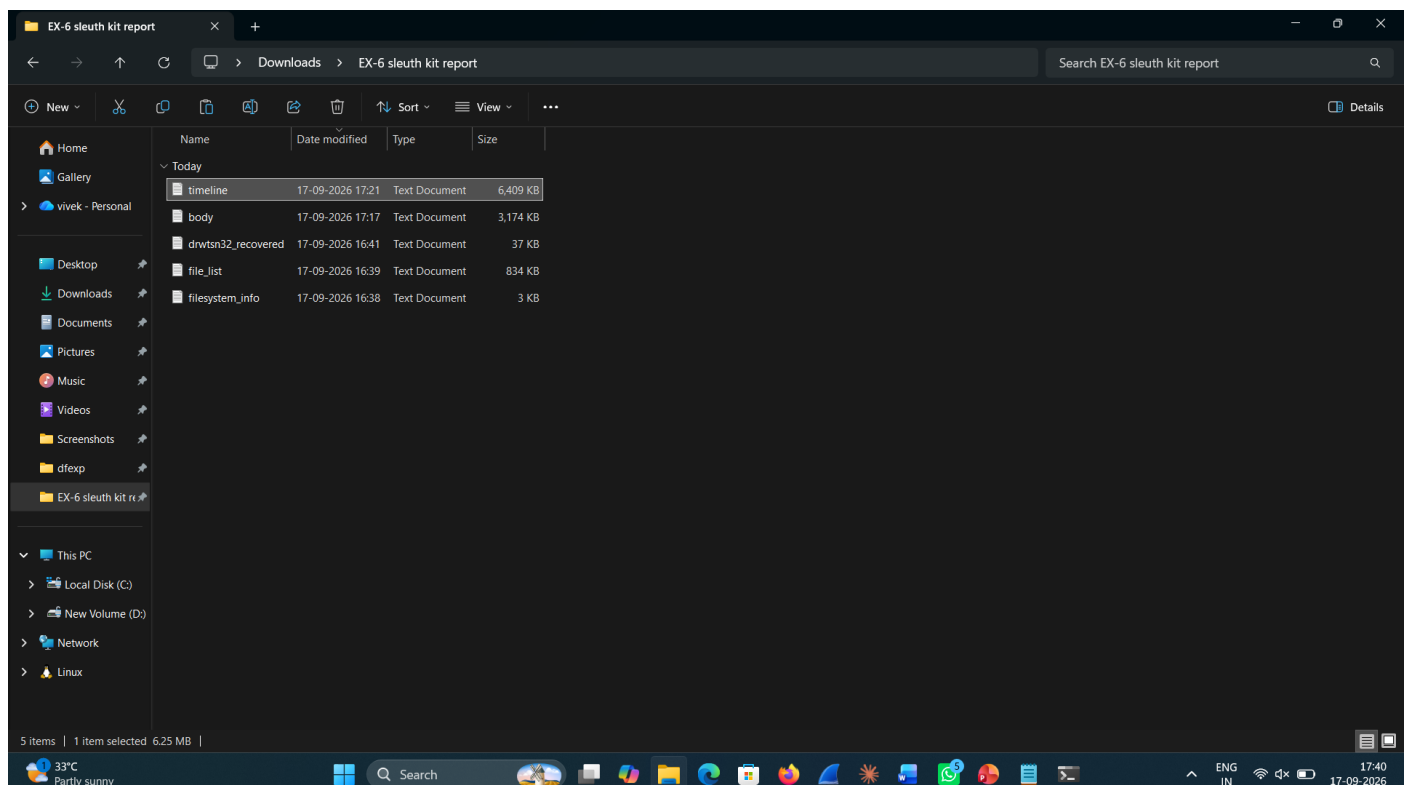
1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The disk image was successfully analysed using Sleuth Kit command-line tools. File systems, partitions, file metadata and deleted files were examined, and relevant forensic information was extracted for further investigation.